



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: Ranjitha R

Reg. No.: 192421285

Course Code: CSA0311

Slot: B

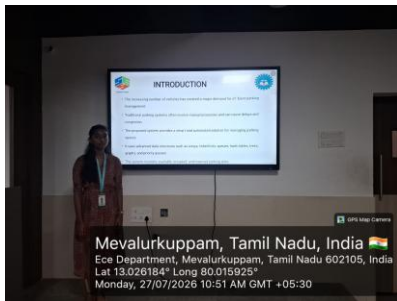
Course Name: DATA STRUCTURE

Course Faculty: Dr. Kiruthika.K

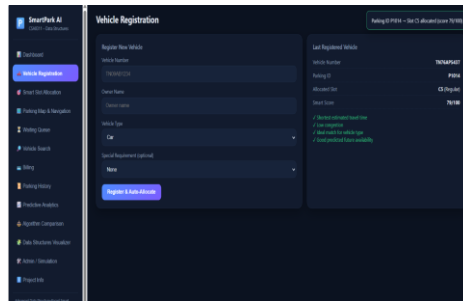
Project Title: Smart Parking Management and Predictive Space Allocation Using Queues and Hash Tables

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

REVIEW 1



MOUDUL OUTPUT



FINAL OUTPUT

ALGORITHM	SEARCH TIME (MS)	Avg WAITING/TOTAL TIME (MIN)	SPACE UTILIZATION
First Available	2.52	2.85	100%
Nearest Slot	4.42	2.85	100%
Priority Queue	7.24	2.85	100%
Proposed Algorithm	7.24	2.85	100%

Project Description: Disease Prediction Module

The Smart Parking Registration and Slot Allocation Module manages the registration of vehicles and the allocation of suitable parking slots. It collects vehicle details such as vehicle number and owner information during entry and generates a unique Parking ID for each registered vehicle for easy identification and tracking. The module checks the real-time availability of parking slots and identifies slots that are available, occupied, or reserved. Based on the current parking status, the system automatically selects and assigns the most suitable available parking slot to the registered vehicle. The assigned slot status is updated from Available → Occupied, while the vehicle and parking details are stored for future reference. This module helps reduce manual parking management, improves slot utilization, and provides an efficient and organized parking allocation system.

Student Signature

Guide Signature